

DAA LAB MCQ's

Lab 1 – Quick Sort

Question No: 1

Multi Choice Type Question

What is the effect of swapping $\text{arr}[i + 1]$ and $\text{arr}[\text{high}]$ in this partition function?

- ☒ It moves the pivot element to its correct position. ✓
- ☐ It places the smallest element in the array at the pivot position.
- ☐ It sorts the array completely.
- ☐ It causes a stack overflow error.

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type	MCQ Single Correct
Blooms taxonomy	Understand								

Question No: 2

Multi Choice Type Question

What will happen if the array is already sorted and the partition function is executed?

- ☐ It will perform optimally with $O(n \log n)$ time complexity.
- ☐ It will result in $O(n^2)$ time complexity, with poor performance.
- ☒ It will work faster than expected due to the sorted state. ✓
- ☐ The partition function will fail to return any result.

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type	MCQ Single Correct
Blooms taxonomy	Understand								

Question No: 3

Multi Choice Type Question

Which of the following scenarios would cause Quick Sort to perform poorly, with a time complexity of $O(n^2)$?

- ☐ When the pivot is always chosen randomly.
- ☒ When the array is already sorted or nearly sorted. ✓
- ☐ When the pivot is selected as the middle element.
- ☐ When the partition function is implemented optimally.

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type	MCQ Single Correct
Blooms taxonomy	Understand								

Question No: 4

Multi Choice Type Question

What is the issue with this code snippet?

```
1 void quickSort(int arr[], int low, int high) {  
2     if (low < high) {  
3         int pivot = arr[low];  
4         int i = low, j = high;  
5         while (i < j) {  
6             while (arr[i] <= pivot) i++;  
7             while (arr[j] > pivot) j--;  
8             if (i < j) {  
9                 int temp = arr[i];  
10                arr[i] = arr[j];  
11                arr[j] = temp;  
12            }  
13        }  
14        arr[low] = arr[j];  
15        arr[j] = pivot;  
16        quickSort(arr, low, j - 1);  
17        quickSort(arr, j + 1, high);  
18    }  
19 }
```

- ☒ The pivot is incorrectly assigned after the loop.
- ☐ The partitioning logic is missing.
- ☐ The array indices are reversed in the partitioning logic.
- ☐ The code doesn't handle empty sub-arrays.

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Medium**

Question type **MCQ Single Correct**

Question No: 5

Multi Choice Type Question

Which of the following is the most efficient way to choose a pivot element in Quick Sort?

☐ Always select the first element as the pivot.

☐ Always select the last element as the pivot.

☒ Randomly select the pivot element.



☐ Always select the middle element as the pivot.

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Medium**

Question type **MCQ Single Correct**

Question No: 6

Multi Choice Type Question

What is the space complexity of the Quick Sort algorithm?

☐ $O(1)$

☐ $O(n)$

☒ $O(\log n)$



☐ $O(n \log n)$

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Medium**

Question type **MCQ Single Correct**

Question No: 7

Multi Choice Type Question

What is the error in the quickSort function?

```
1 void quickSort(int arr[], int low, int high) {  
2     if (low < high) {  
3         int pivotIndex = (low + high) / 2;  
4         int pivot = arr[pivotIndex];  
5         // Swap pivot with the last element  
6         int temp = arr[pivotIndex];  
7         arr[pivotIndex] = arr[high];  
8         arr[high] = temp;  
9         quickSort(arr, low, high - 1);  
10        quickSort(arr, pivotIndex + 1, high);  
11    }  
12 }  
13 }
```

- ☐ The pivot index calculation is incorrect.
- ☐ The recursive calls should be swapped.
- ☒ The pivot value isn't correctly placed.
- ☐ The pivot element is chosen incorrectly.

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 8

Multi Choice Type Question

Why might using $\text{arr}[(\text{low} + \text{high})/2]$ as the pivot cause issues in some Quick Sort implementations?

- ☐ It results in an infinite loop.
- ☐ It is inefficient because the pivot is always the middle element.
- ☒ It does not guarantee the array will be partitioned correctly.
- ☐ It causes the function to return a null value.

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 9

Multi Choice Type Question

What will be the output if this partition function is executed on the array {10, 7, 8, 9, 1, 5} with low = 0 and high = 5?

```
1 int partition(int arr[], int low, int high) {  
2     int pivot = arr[high];  
3     int i = low;  
4     for (int j = low; j < high; j++) {  
5         if (arr[j] < pivot) {  
6             swap(arr, i, j);  
7             i++;  
8         }  
9     }  
10    swap(arr, i, high);  
11    return i;  
12 }  
13 }
```

☐ 5

☒ 3 ✓

☐ 2

☐ 4

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Data Structure**

Question No: 10

Multi Choice Type Question

What is the worst-case time complexity of the Quick Sort algorithm?

☐ $O(n \log n)$

☐ $O(n)$

☒ $O(n^2)$

☐ $O(\log n)$

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 11

Multi Choice Type Question

What do the 'low' and 'high' parameters represent in this code?

```
1 void quickSort(int arr[], int low, int high) {  
2     if (low < high) {  
3         int pivot = partition(arr, low, high);  
4         quickSort(arr, low, pivot - 1);  
5         quickSort(arr, pivot + 1, high);  
6     }  
7 }
```

- ☐ The number of comparisons performed
- ☒ The range of elements to sort within the array
- ☐ The value of the pivot element
- ☐ The number of swaps made during sorting

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 12

Multi Choice Type Question

In a quick sort algorithm, where are the smaller elements placed to the pivot during the partition process, assuming we are sorting in increasing order?

- ☐ To the right of the pivot
- ☒ To the left of the pivot
- ☐ The pivot element will be removed
- ☐ None of the mentioned options

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Data Structure** | Topic **Sorting** | Sub Topic **Quick sort**

Question No: 13

Multi Choice Type Question

What is the error in the given code?

```
1 void quickSort(int arr[], int low, int high) {  
2     if (low < high) {  
3         int pivot = partition(arr, low, high);  
4         quickSort(arr, pivot + 1, high);  
5         quickSort(arr, low, pivot - 1);  
6     }  
7 }  
8
```

- ☐ The partition function is missing.
- ☒ The recursive calls for quickSort are incorrectly swapped.
- ☐ The base case for recursion is missing.
- ☐ The variable pivot is not declared.

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 14

Multi Choice Type Question

Why is Quick Sort considered faster than Merge Sort on average, despite both having the same time complexity in the best case?

- ☒ Quick Sort uses less memory due to being an in-place algorithm. ✓
- ☐ Merge Sort always requires additional space for merging.
- ☐ Quick Sort avoids recursive calls.
- ☐ Quick Sort guarantees $O(n \log n)$ time in every case.

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Data Structure**

Multi Choice Type Question

What will happen if the recursive calls in this quickSort function are swapped?

```
1 void quickSort(int arr[], int low, int high) {  
2     if (low < high) {  
3         int pivot = partition(arr, low, high);  
4         quickSort(arr, pivot + 1, high);  
5         quickSort(arr, low, pivot - 1);  
6     }  
7 }  
8
```

- ☐ It will cause infinite recursion.
- ☐ It will cause the array to be sorted in reverse order.
- ☒ It will still sort the array correctly.
- ☐ It will cause the function to terminate early.

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type	MCQ Single Correct
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Lab 2 – Merge Sort and Binary Search

Question No: 1

Multi Choice Type Question

Which of the following statements is true about the merge sort algorithm?

- ☐ It is not a stable sorting algorithm
- ☒ It requires additional memory for merging
- ☐ It is not suitable for sorting large datasets
- ☐ None of the mentioned options

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 2

Multi Choice Type Question

Suppose you are using Binary Search on a sorted array of even length (e.g., 8 elements). At each step, you calculate the middle as:

$\text{mid} = (\text{low} + \text{high}) // 2$

Which of the following statements is always true?

- ☐ The mid index always divides the array exactly into two equal halves.
- ☐ The mid index will always be closer to low than to high.
- ☐ The element at the mid index will be the median.
- ☒ The mid index may not divide the array into equal halves in all steps. ✓

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Data Structure**

Question No: 3

Multi Choice Type Question

In a recursive implementation of Binary Search, if the update conditions are not carefully written, an infinite loop may occur. Which of the following update conditions is likely to cause an infinite loop?

- ☐ if `arr[mid] < x: low = mid + 1`
- ☐ if `arr[mid] > x: high = mid`
- ☒ if `arr[mid] < x: low = mid` ✓
- ☐ if `arr[mid] == x: return mid`

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Data Structure** | Topic **Searching** | Sub Topic **Binary Search** | Blooms taxonomy **Analyse**

Question No: 4

Multi Choice Type Question

Given a sorted array of 1000 unique integers, how many maximum comparisons does Binary Search make in the worst case?

- ☒ 10 ✓
- ☐ 1000
- ☐ 500
- ☐ 999

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Data Structure**

Question No: 5

Multi Choice Type Question

While merging two sorted arrays $A = [3, 6]$ and $B = [2, 5]$, which of the following is the correct order of comparisons during the merge process?

- ☐ 3 vs 2 → 6 vs 5
- ☐ 3 vs 5 → 6 vs 2
- ☐ 2 vs 3 → 5 vs 6
- ☒ 3 vs 2 → 3 vs 5 → 6 vs 5 ✓

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Data Structure** | Topic **Sorting**

Question No: 6

Multi Choice Type Question

What is the role of this code in merge()?

```
1 while (i < n1) {  
2     arr[k] = L[i];  
3     i++;  
4     k++;  
5 }  
6 while (j < n2) {  
7     arr[k] = R[j];  
8     j++;  
9     k++;  
10 }
```

☐ Merge only the smallest values

☐ Merge duplicate values only

☒ Copy any remaining elements after merging ✓

☐ Count elements from both arrays

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Data Structure**

Question No: 7

Multi Choice Type Question

You apply Binary Search on the sorted array [3, 7, 10, 15, 21, 30] to find the element 13. Which sequence of middle elements will Binary Search compare with before declaring that the element is not present?

☐ 15 → 7 → 10

☒ 10 → 21 → 15 ✓

☐ 10 → 15 → 21

☐ 21 → 15 → 10

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Data Structure** | Topic **Searching** | Sub Topic **Binary Search** | Blooms taxonomy **Analyse**

Question No: 8

Multi Choice Type Question

If Merge Sort is implemented iteratively instead of recursively, what is the primary change in approach?

☒ Use of a loop instead of recursion for merging subarrays

☐ The need for a different sorting algorithm

☐ Increased space complexity

☐ It changes the time complexity to $O(n^2)$

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 9

Multi Choice Type Question

A student implements Binary Search and calculates the middle index as:

$mid = (low + high) // 2$

But when the input values of low and high are both large (e.g., $low = 2_000_000_000$, $high = 2_000_000_010$), the program crashes due to integer overflow. What is the best fix?

☐ Use $mid = (low + high) / 2.0$ to avoid overflow

☐ Use $mid = (low + high) // 2$ in all cases

☒ Use $mid = low + (high - low) // 2$

☐ There is no way to avoid overflow

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Data Structure** | Topic **Searching** | Sub Topic **Binary Search**

Question No: 10

Multi Choice Type Question

You are using Binary Search to find the first occurrence of a value x in a sorted array that may contain duplicates. Which of the following modifications is necessary?

☐ Use linear search before applying Binary Search.

☐ Always move low to $mid + 1$ if element is found.

☒ When $x == arr[mid]$, move high to $mid - 1$.

☐ Binary Search cannot be used when duplicates exist.

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Data Structure** | Topic **Searching** | Sub Topic **Binary Search**

Question No: 11

Multi Choice Type Question

How many comparisons will be made during the merging step when merging two sorted sublists of 4 elements each?

☐ At most 4

☒ Exactly 7



☐ At most 8

☐ At least 10

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Easy	Question type	MCQ Single Correct	Subject	Data Structure
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Question No: 12

Multi Choice Type Question

What happens during the merge step in Merge Sort?

☒ Two sorted subarrays are combined into one sorted array

☐ The array is divided into smaller subarrays

☐ Elements are swapped to bring the smallest element to the front

☐ The array is rearranged using the pivot element

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type	MCQ Single Correct
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Question No: 13

Multi Choice Type Question

If a Merge Sort is applied to the list 25, 10, 15, 5, what is the result after the first merge pass (i.e., after first level merging)?

☐ 5, 10, 15, 25

☒ 10, 25, 5, 15



☐ 10, 25, 15, 5

☐ 25, 10, 15, 5

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Data Structure**

Question No: 14

Multi Choice Type Question

Given 34, 8, 50, 20, what is the state of the array after second level merging in Merge Sort?

☐ 8, 34, 20, 50

☒ 8, 20, 34, 50

☐ 8, 34, 50, 20

☐ 20, 34, 8, 50

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Multi Choice Type Question

You are given a sorted array that is rotated at some unknown pivot.

Example: [15, 18, 2, 3, 6, 12]

You need to find if a target value 3 exists using Binary Search logic. Which of the following is the correct key idea to modify Binary Search in this case?

- ☐ Always check both halves recursively, since the array is not sorted.
- ☒ Determine which half is sorted and discard the unsorted half accordingly. ✓
- ☐ Sort the array first, then apply standard Binary Search.
- ☐ Apply linear search, as Binary Search won't work on rotated arrays.

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Easy	Question type	MCQ Single Correct	Subject	Data Structure	Topic	Searching
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Lab 3 – Prim's and Kruskal

Question No: 1

Multi Choice Type Question

What is the primary difference between Prim's and Kruskal's algorithm in terms of how edges are selected?

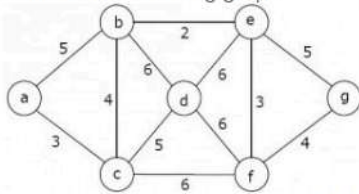
- ☐ Both algorithms are identical in edge selection strategy.
- ☐ Kruskal's algorithm starts from an arbitrary vertex, while Prim's algorithm starts from any vertex in the graph.
- ☒ Prim's algorithm grows the MST from a starting vertex by always adding the minimum weight edge connecting the MST to a non-MST vertex, while Kruskal's algorithm considers all edges globally and adds the minimum weight edge that doesn't create a cycle. ✓
- ☐ Prim's algorithm selects edges based on vertex connectivity, while Kruskal's algorithm selects edges based on their weights.

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type	MCQ Single Correct	Subject
Program Outcome	PO1									

Question No: 2

Multi Choice Type Question

Consider the following graph:



Which one of the following is NOT the sequence of edges added to the minimum spanning tree using Kruskal's algorithm?

☐ (b,e)(e,f)(a,c)(b,c)(f,g)(c,d)

☐ (b,e)(e,f)(a,c)(f,g)(b,c)(c,d)

☐ (b,e)(a,c)(e,f)(b,c)(f,g)(c,d)

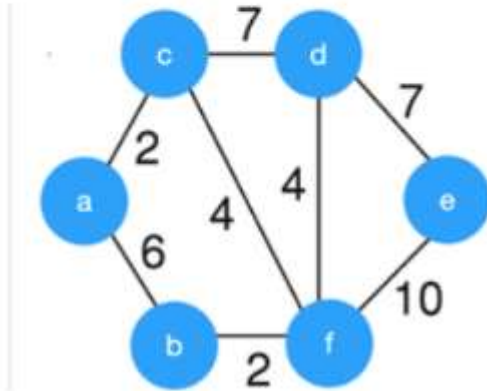
☒ (b,e)(e,f)(b,c)(a,c)(f,g)(c,d) ✓

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type	MCQ Single Correct	Subject
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Question No: 3

Multi Choice Type Question

Consider the given graph.



What is the weight of the minimum spanning tree using Kruskal's algorithm?

☐ 24

☐ 23

☐ 15

☒ 19

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Medium**

Question No: 4

Multi Choice Type Question

In which scenario would Kruskal's algorithm perform better than Prim's algorithm?

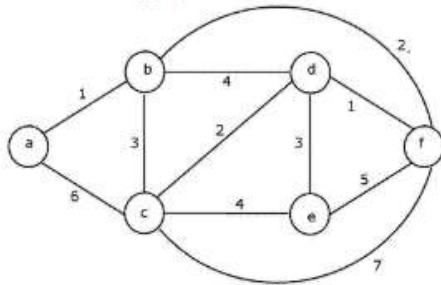
- ☐ When the graph is dense
- ☒ When the graph is sparse
- ☐ When the graph is a complete graph
- ☐ When there is no cycle in the graph

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type
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Question No: 5

Multi Choice Type Question

Consider the following graph:



Which one of the following cannot be the sequence of edges added, in that order, to a minimum spanning tree using Kruskal's algorithm?

☐ (a-b), (d-f), (b-f), (d-c), (d-e)

☐ (a-b), (d-f), (d-c), (b-f), (d-e)

☐ (d-f), (a-b), (d-c), (b-f), (d-e)

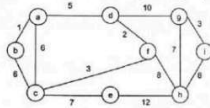
☒ (d-f), (a-b), (b-f), (d-e), (d-c)

Status **Correct** Mark obtained **1/1** Hints used **0** Level **Medium** Question type **MCQ Single Correct** Subject **Algorithms** Topic

Question No: 6

Multi Choice Type Question

For the undirected, weighted graph given below, which of the following sequences of edges represents a correct execution of Prim's algorithm to construct a Minimum Spanning Tree?



☐ (a, b), (d, f), (f, c), (g, i), (d, a), (g, h), (c, e), (f, h)

☐ (c, e), (c, f), (f, d), (d, a), (a, b), (g, h), (h, f), (g, i)

☒ (d, f), (f, c), (d, a), (a, b), (c, e), (f, h), (g, h), (g, i)

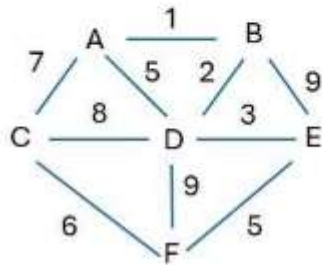
☐ (h, g), (g, i), (h, f), (f, c), (f, d), (d, a), (a, b), (c, e)

Status **Correct** Mark obtained **1/1** Hints used **0** Level **Easy** Question type **MCQ Single Correct** Subject **Algorithms** Topic **Greedy** Sub Topic **Greedy** Blooms taxonomy

Question No: 7

Multi Choice Type Question

Consider the following graph:



Which edges would be included in the minimum spanning tree using Prim's algorithm starting from vertex A?

☒ AB, BD, DE, EF, FC

☐ AC, CD, DE, EB, BF

☐ AB, BD, DE, EC, CF

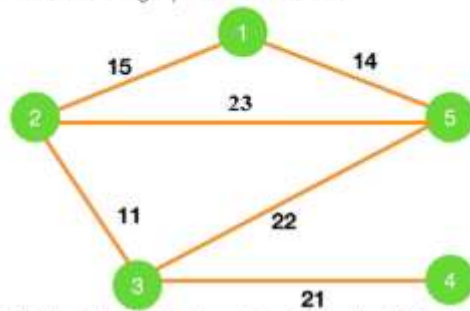
☐ AC, CD, DE, EB, FE

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Question No: 8

Multi Choice Type Question

Consider the graph shown below.



Which of the following edges form the MST of the given graph using Prim's algorithm, starting from vertex 4.

☐ (4-3)(5-3)(2-3)(1-2)

☐ (4-3)(3-5)(5-1)(1-2)

☐ (4-3)(3-5)(5-2)(1-5)

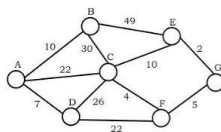
☒ (4-3)(3-2)(2-1)(1-5)

Status **Correct** Mark obtained **1/1** Hints used **0** Level **Easy** Question type **MCQ Single Correct**

Question No: 9

Multi Choice Type Question

Consider the undirected graph below:



Using Prim's algorithm to construct a minimum spanning tree starting with node A, which one of the following sequences of edges represents a possible order in which the edges would be added to construct the minimum spanning tree?

☐ (A, B), (A, D), (D, F), (F, G), (G, E), (F, C)

☐ (E, G), (C, F), (F, G), (A, D), (A, B), (A, C)

☐ (A, D), (A, B), (A, C), (C, F), (G, E), (F, G)

☒ (A, D), (A, B), (D, F), (F, C), (F, G), (G, E) ✓

Status **Correct** Mark obtained **1/1** Hints used **0** Level **Easy** Question type **MCQ Single Correct** Subject **Programming** Topic **Algorithms** Sub Topic **Greedy Algorithms** Blooms taxonomy **Analyse**

Question No: 10

Multi Choice Type Question

Kruskal's algorithm is used to _____.

☒ Find minimum spanning tree ✓

☐ Find single source shortest path

☐ Find all pair shortest path algorithm

☐ Traverse the graph

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Algorithms** | Topic

Question No: 11

Multi Choice Type Question

Prim's algorithm is a _____.

☐ Divide and conquer algorithm

☒ Greedy algorithm

☐ Dynamic Programming

☐ Approximation algorithm

Status **Correct** | Mark obtained **1/1** | Hints used **0**

Question No: 12

Multi Choice Type Question

Which of the following is true about the edges selected by Kruskals algorithm?

- ☐ It always selects the edge with the smallest weight that connects two unvisited vertices.
- ☐ It always selects the edge with the largest weight that does not form a cycle.
- ☒ It sorts the edges in non-decreasing order of weights and selects them one by one.
- ☐ It starts by selecting the edges with the highest number of common vertices.

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 13

Multi Choice Type Question

In Kruskals algorithm, how do we ensure no cycles are formed when adding edges to the MST?

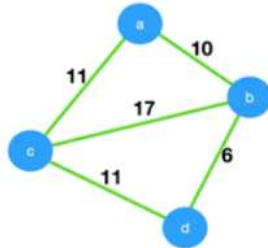
- ☐ By using a max-heap to store edges
- ☒ By using a union-find data structure
- ☐ By sorting edges based on vertices
- ☐ By visiting all nodes in a depth-first search order

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 14

Multi Choice Type Question

Consider the given graph.



What is the weight of the minimum spanning tree using Prim's algorithm, starting from vertex a?

☐ 23

☐ 28

☒ 27



☐ 11

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Data Structure**

Question No: 15

Multi Choice Type Question

Which of the following is true?

- ☒ Prim's algorithm initializes with a vertex
- ☐ Prim's algorithm initializes with a edge
- ☐ Prim's algorithm initializes with a vertex which has smallest edge
- ☐ Prim's algorithm initializes with a forest

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type

Lab 4 –Dijkstra

Question No: 1

Multi Choice Type Question

What is the time complexity of Dijkstra's algorithm for finding the shortest path in a graph with V vertices and E edges?

- ☐ $O(V)$
- ☐ $O(E)$
- ☐ $O(V \log V)$
- ☒ $O(E \log V)$ ✓

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Data Structure**

Question No: 2

Multi Choice Type Question

In Dijkstra's algorithm, what happens when a shorter path to a neighboring vertex is found?

- ☐ The neighbor vertex is removed from the graph
- ☒ The tentative distance of the neighbor vertex is updated ✓

☐ The algorithm terminates immediately

☐ The current vertex is skipped

Status: **Correct** | Mark obtained: **1/1** | Hints used: **0** | Level: **Medium** | Question type: **MCQ Single Correct** | Subject: **Data Structure**

Question No: 3

Multi Choice Type Question

In Dijkstra's algorithm, what is the name of the data structure used to keep track of the shortest path to each vertex?

☒ Priority Queue ✓

☐ Queue

☐ Hash Table

☐ Stack

Status: **Correct** | Mark obtained: **1/1** | Hints used: **0** | Level: **Medium** | Question type: **MCQ Single Correct** | Subject: **Data Structure**

Question No: 4

Multi Choice Type Question

Which data structure modification can help improve Dijkstra's algorithm performance on dense graphs?

☒ Using a Fibonacci heap instead of a binary heap ✓

☐ Using a queue instead of a heap

☐ Using a stack instead of a queue

☐ Using a linked list instead of an array

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type	MCQ Single Correct	Blooms taxonomy	Remember
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Question No: 5

Multi Choice Type Question

In Dijkstra's algorithm, what type of edge weights are allowed?

☒ Positive and zero edge weights

☐ Only positive edge weights

☐ Only negative edge weights

☐ Any type of edge weights

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Easy
--------	----------------	---------------	------------	------------	----------	-------	-------------

Question No: 6

Multi Choice Type Question

If a graph contains negative weight edges, what will happen when running Dijkstra's algorithm?

- ☐ It will compute the correct shortest paths
- ☒ It may fail to compute correct shortest paths
- ☐ It will detect and report negative cycles
- ☐ It will run faster

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 7

Multi Choice Type Question

What does Dijkstra's algorithm use to make its decisions at each step of the process?

- ☐ Breadth-first search
- ☐ Random choice
- ☒ Greedy approach
- ☐ Depth-first search

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Question No: 8

Multi Choice Type Question

What is the initial tentative distance assigned to the source vertex in Dijkstra's algorithm?

☐ Infinity

☒ Zero

☐ One

☐ Undefined

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 9

Multi Choice Type Question

Which of the following algorithms can be used to handle graphs with negative edge weights where Dijkstra's algorithm fails?

☒ Bellman-Ford algorithm



☐ Prim's algorithm

☐ Kruskal's algorithm

☐ Depth-first search

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Data Structure**

Question No: 10

Multi Choice Type Question

After the execution of Dijkstra's algorithm, what information do we get for each vertex?

- ☐ The number of edges connected to it
- ☒ The shortest distance from the source vertex and the path taken
- ☐ The longest path from the source vertex
- ☐ The average weight of incident edges

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 11

Multi Choice Type Question

Which of the following is not a step in Dijkstra's algorithm?

- ☐ Assign all vertices a tentative distance value.
- ☐ Choose the vertex with the shortest tentative distance value.
- ☐ Update the distance value of adjacent vertices to the chosen vertex.
- ☒ Remove the chosen vertex from the graph.

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium**

Question No: 12

Multi Choice Type Question

Which of the following is a common way to implement the priority queue in Dijkstra's algorithm for better efficiency?

☐ Using an array

☐ Using a linked list

☒ Using a binary heap



☐ Using a stack

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Data Structure**

Question No: 13

Multi Choice Type Question

How does Dijkstra's algorithm handle cycles in a graph?

☐ It cannot handle cycles

☐ It ignores cycles

☒ It works correctly if edge weights are non-negative

☐ It only processes acyclic graphs

Status **Correct** | Mark obtained **1/1** | Hints used **0**

Question No: 14

Multi Choice Type Question

Which of these graph representations is most suitable for implementing Dijkstra's algorithm efficiently?

☒ Adjacency list

☐ Adjacency matrix

☐ Edge list

☐ Incidence matrix

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 15

Multi Choice Type Question

What is the role of the "visited" or "finalized" set in Dijkstra's algorithm?

☐ To store all nodes in the graph

☒ To keep track of nodes whose shortest distance is finalized

☐ To store all edges traversed

☐ To store nodes to be processed next

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium**

Lab 5 – Matrix Multiplications

Question No: 1

Multi Choice Type Question

What is the time complexity of the Matrix Chain Multiplication algorithm using dynamic programming?

☐ $O(n \log n)$

☐ $O(n^4)$

☒ $O(n^3)$

☐ $O(n^2)$

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Easy	Question type	MCQ Single Correct
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Question No: 2

Multi Choice Type Question

How many recursive calls will be made when executing this code?

```
1 class MCM {  
2     static int count = 0;  
3     static int mcm(int[] arr, int i, int j) {  
4         count++;  
5         if (i == j) return 0;  
6         int min = Integer.MAX_VALUE;  
7         for (int k = i; k < j; k++) {  
8             int cost = mcm(arr, i, k) + mcm(arr, k+1, j) + arr[i-1]*arr[k]*arr[j];  
9             min = Math.min(min, cost);  
10        }  
11        return min;  
12    }  
13    public static void main(String[] args) {  
14        int arr[] = {10, 20, 30, 40};  
15        mcm(arr, 1, arr.length-1);  
16        System.out.println(count);  
17    }  
18 }
```

☐ 10

☐ 8

☒ 9

☐ 5

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Question No: 3

Multi Choice Type Question

Which approach is generally preferred for solving the Matrix Chain Multiplication problem for large numbers of matrices?

☐ Brute-force approach

☒ Dynamic Programming

☐ Greedy algorithm

☐ Divide and Conquer

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Algorithms**

Question No: 4

Multi Choice Type Question

What will be the output of the following code?

```
1 class MCM {
2     static int mcm(int p[], int i, int j) {
3         if (i == j) return 0;
4         int min = Integer.MAX_VALUE;
5         for (int k = i; k < j; k++) {
6             int cost = mcm(p, i, k) + mcm(p, k+1, j) + p[i-1]*p[k]*p[j];
7             if (cost < min) min = cost;
8         }
9         return min;
10    }
11    public static void main(String[] args) {
12        int arr[] = {10, 20, 30};
13        System.out.println(mcm(arr, 1, arr.length-1));
14    }
15 }
```

☒ 6000

☐ 10000

☐ 9000

☐ 5000

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Question No: 5

Multi Choice Type Question

If there are 4 matrices to be multiplied, how many ways can they be parenthesized?

☐ 6

☒ 5

☐ 14

☐ 4

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Question No: 6

Multi Choice Type Question

What is the parenthesization strategy in Matrix Chain Multiplication?

- ☐ To multiply matrices from left to right
- ☐ To multiply matrices from right to left
- ☒ To explore all possible ways to parenthesize and choose the optimal one
- ☐ To divide the matrices into equal halves and multiply

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Question No: 7

Multi Choice Type Question

What is the primary objective of the Matrix Chain Multiplication problem?

- ☐ To multiply matrices in sequential order.
- ☐ To maximize the number of scalar multiplications.
- ☒ To minimize the number of scalar multiplications.
- ☐ To find the determinant of the matrices.

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type

Question No: 8

Multi Choice Type Question

What will be the output of the following code?

```
1 class MCM {
2     static int matrixChainOrder(int p[], int n) {
3         int m[][] = new int[n][n];
4         for (int L = 2; L < n; L++) {
5             for (int i = 1; i < n - L + 1; i++) {
6                 int j = i + L - 1;
7                 m[i][j] = Integer.MAX_VALUE;
8                 for (int k = i; k <= j - 1; k++) {
9                     int q = m[i][k] + m[k+1][j] + p[i-1]*p[k]*p[j];
10                    if (q < m[i][j]) m[i][j] = q;
11                }
12            }
13        }
14        return m[1][n-1];
15    }
16    public static void main(String args[]) {
17        int arr[] = {40, 20, 30, 10, 30};
18        System.out.println(matrixChainOrder(arr, arr.length));
19    }
20 }
21 }
```

☒ 26000

☐ 30000

☐ 32000

☐ 60000

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type

Question No: 9

Multi Choice Type Question

In the Matrix Chain Multiplication problem, what is the significance of splitting the chain at position k ?

- ☐ It helps to find the matrix with the least number of rows.
- ☒ It determines the optimal order of matrix multiplication.
- ☐ It divides the chain into equal halves.
- ☐ It ensures that no matrix is multiplied more than once.

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Easy	Question type	MCQ Single Correct
--------	---------	---------------	-----	------------	---	-------	------	---------------	--------------------

Question No: 10

Multi Choice Type Question

Which of the following is used to solve the Matrix Chain Multiplication problem efficiently?

- ☐ Backtracking
- ☒ Dynamic Programming
- ☐ Divide and Conquer
- ☐ Greedy Algorithm

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Easy	Question type	MCQ Single Correct
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Question No: 11

Multi Choice Type Question

What is the minimum number of matrices required for Matrix Chain Multiplication to make sense?

☐ 1

☒ 2

☐ 3

☐ 4

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Easy**

Question type **MCQ Single Correct**

Question No: 12

Multi Choice Type Question

What will be the minimal cost of multiplying these matrices?

```
1 public class MCM {  
2     static int mcm(int[] arr, int i, int j) {  
3         if(i == j) return 0;  
4         int min = Integer.MAX_VALUE;  
5         for(int k = i; k < j; k++) {  
6             int cost = mcm(arr, i, k) + mcm(arr, k+1, j) + arr[i-1]*arr[k]*arr[j];  
7             min = Math.min(min, cost);  
8         }  
9         return min;  
10    }  
11    public static void main(String[] args) {  
12        int arr[] = {5, 10, 3, 12};  
13        System.out.println(mcm(arr, 1, arr.length-1));  
14    }  
15 }  
16
```

☐ 270

☒ 330

☐ 390

☐ 200

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 13

Multi Choice Type Question

In Matrix Chain Multiplication, which of the following represents the recursive subproblem?

☐ Multiplying matrices from index i to j

☐ Calculating the matrix product of matrices i and j

☒ Finding the minimum number of scalar multiplications for a matrix chain

☐ Storing the matrix dimensions

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Question No: 14

Multi Choice Type Question

What will be the output of the following code?

```
1 class MCM {
2     static int mcm(int[] arr, int i, int j) {
3         if(i == j) return 0;
4         int min = Integer.MAX_VALUE;
5         for(int k = i; k < j; k++) {
6             int cost = mcm(arr, i, k) + mcm(arr, k+1, j) + arr[i-1]*arr[k]*arr[j];
7             min = Math.min(min, cost);
8         }
9         return min;
10    }
11    public static void main(String[] args) {
12        int arr[] = {5, 10, 3, 12};
13        System.out.println(mcm(arr, 1, arr.length-1));
14    }
15 }
```

☐ 144

☒ 156

☐ 189

☐ 200

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Question No: 15

Multi Choice Type Question

If there are n matrices, what is the size of the dimensions array used in Matrix Chain Multiplication?

☐ 2n

☐ n-1

☒ n+1

☐ n

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Lab 6 – Optimal BST

Question No: 1

Multi Choice Type Question

A node of the weight-balanced tree has

- ☐ Key, size
- ☐ Key, value
- ☒ Key, left and right pointers, size
- ☐ Key

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Question No: 2

Multi Choice Type Question

What is the primary benefit of an OBST in practical applications?

- ☐ It eliminates the need for rebalancing
- ☒ It reduces the average search time compared to a regular BST
- ☐ It can handle infinite keys efficiently
- ☐ It always results in a perfectly balanced tree

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Easy**

Question No: 3

Multi Choice Type Question

What is the primary objective of constructing an Optimal Binary Search Tree (OBST)?

- ☐ To ensure all elements are equally distributed
- ☒ To minimize the expected search cost based on access probabilities
- ☐ To maximize the height of the tree
- ☐ To minimize the number of nodes in the tree

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type

Question No: 4

Multi Choice Type Question

What are the applications of a weight-balanced tree?

☒ Dynamic sets, dictionaries, sequences, and maps

☐ Heaps

☐ Sorting

☐ Storing strings

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Easy**

Question No: 5

Multi Choice Type Question

Which of the following is NOT a property of an Optimal Binary Search Tree?

- ☐ It considers access probabilities for optimal structure
- ☐ It is computed using a dynamic programming approach
- ☒ It is always a balanced binary search tree
- ☐ It minimizes the expected search cost

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type

Question No: 6

Multi Choice Type Question

What do the below definitions convey?

- i. A binary tree is balanced if, for every node, it is going to hold that the number of inner nodes in the left subtree and the number of inner nodes in the right subtree differ by at most 1.
- ii. A binary tree is balanced if, for any two leaves, the difference in depth is at most 1.

- ☐ Definitions of height-balanced tree
- ☐ Definitions of weight-balanced trees
- ☐ Height-balanced and weight-balanced tree definitions
- ☒ Weight-balanced and height-balanced tree definitions ✓

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Data Structure** | Topic **Tree** | Sub Topic **Binary Search Tree**

Question No: 7

Multi Choice Type Question

What is a weight-balanced tree?

- ☒ A binary tree that stores the sizes of subtrees in nodes
- ☐ A binary tree with an additional attribute of weight
- ☐ A height balanced binary tree
- ☐ A normal binary tree

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Easy**

Question No: 8

Multi Choice Type Question

In an OBST, what additional information is required apart from the keys and their values?

- ☐ Pre-order traversal sequence
- ☐ Hash values of keys
- ☐ Parent-child relationships
- ☒ Frequency of access (probabilities) ✓

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Question No: 9

Multi Choice Type Question

Which algorithm is commonly used to construct an Optimal Binary Search Tree?

- ☐ Backtracking
- ☐ Divide and Conquer
- ☒ Dynamic Programming
- ☐ Greedy Algorithm

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type

Question No: 10

Multi Choice Type Question

When constructing an OBST using dynamic programming, what does the recurrence relation primarily depend on?

- ☐ The in-order traversal of the tree
- ☐ The depth-first search traversal of the tree
- ☐ The number of keys present in the tree
- ☒ The left and right subtrees cumulative search cost ✓

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject

Question No: 11

Multi Choice Type Question

What are the operations that can be performed on weight balanced tree?

- ☒ All basic operations and set intersection, set union and subset tests
- ☐ All basic operations
- ☐ Only insertion and deletion
- ☐ Set intersection, set union and subset test

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** |

Question No: 12

Multi Choice Type Question

The cost function of an OBST is computed based on which factor?

- ☐ The number of edges in the tree
- ☐ The in-order traversal of the tree
- ☒ The depth of nodes weighted by their probabilities
- ☐ The number of child nodes

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy**

Question No: 13

Multi Choice Type Question

In the dynamic programming approach for OBST, which table is used to store the computed minimum cost for subproblems?

- ☐ Adjacency matrix
- ☐ Probability matrix
- ☒ Cost matrix ✓
- ☐ Prefix sum table

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct** | Subject **Data Structure**

Question No: 14

Multi Choice Type Question

What is the main difference between a regular Binary Search Tree (BST) and an Optimal Binary Search Tree (OBST)?

- ☐ OBST requires more space than BST
- ☒ OBST minimizes the search cost, considering probabilities of access, whereas BST does not ✓
- ☐ OBST is balanced, whereas BST is not
- ☐ OBST allows duplicate keys, while BST does not

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Easy	Question type	MCQ Single Correct	Subject
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Question No: 15

Multi Choice Type Question

What happens if the frequency of some keys in an OBST is significantly higher than others?

- ☐ The tree will become a complete binary tree
- ☐ The search cost will increase
- ☐ The tree will always remain balanced
- ☒ The tree will become unbalanced ✓

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Easy	Question type	MCQ Single Correct	Subject
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Lab 7 – LCS

Question No: 1

Multi Choice Type Question

What will be the output of the following code for the input strings "AGGTAB" and "GTXAYB"?

```
1 public class LCS {  
2     public static void main(String[] args) {  
3         String s1 = "AGGTAB";  
4         String s2 = "GTXAYB";  
5         int m = s1.length(), n = s2.length();  
6         int[][] dp = new int[m+1][n+1];  
7  
8         for (int i = 0; i <= m; i++) {  
9             for (int j = 0; j <= n; j++) {  
10                if (i == 0 || j == 0)  
11                    dp[i][j] = 0;  
12                else if (s1.charAt(i-1) == s2.charAt(j-1))  
13                    dp[i][j] = dp[i-1][j-1] + 1;  
14                else  
15                    dp[i][j] = Math.max(dp[i-1][j], dp[i][j-1]);  
16            }  
17        }  
18        System.out.println(dp[m][n]);  
19    }  
20 }
```

☒ 4

☐ 3

☐ 5

☐ 6

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type
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Question No: 2

Multi Choice Type Question

Which dynamic programming approach can be used to solve the Longest Increasing Subsequence (LIS) problem in $O(n \log n)$ time complexity?

☐ Bottom-Up Approach

☐ Top-Down Approach

☒ Binary Indexed Tree (BIT) ✓

☐ Segment Tree

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Programming** | Topic **Algorithms**
Blooms taxonomy **Analyse**

Question No: 3

Multi Choice Type Question

Which base case is correct for LCS DP?

☐ $dp[0][j] = -1$ and $dp[i][0] = -1$

☐ $dp[0][j] = 1$ and $dp[i][0] = 1$

☐ $dp[0][j] = j$ and $dp[i][0] = i$

☒ $dp[0][j] = 0$ and $dp[i][0] = 0$

Status **Correct** | Mark obtained **1/1** | Hints used **0**
Blooms taxonomy **Apply**

Question No: 4

Multi Choice Type Question

Find the longest increasing subsequence for the given sequence: {10, -10, 12, 9, 10, 15, 13, 14}

☐ {10, 12, 15}

☐ {10, 12, 13, 14}

☐ {-10, 12, 13, 14}

☒ {-10, 9, 10, 13, 14}



Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Medium**

Question type **MCQ Single Correct**

Blooms taxonomy **Analyse**

Multi Choice Type Question

What will be the output of the following code?

```
1 class LCS {
2     public static void main(String[] args) {
3         String s1 = "";
4         String s2 = "ABCD";
5         System.out.println(lcs(s1, s2));
6     }
7
8     static int lcs(String s1, String s2) {
9         int m = s1.length();
10        int n = s2.length();
11        int[][] dp = new int[m + 1][n + 1];
12
13        for (int i = 1; i <= m; i++) {
14            for (int j = 1; j <= n; j++) {
15                if (s1.charAt(i - 1) == s2.charAt(j - 1)) {
16                    dp[i][j] = dp[i - 1][j - 1] + 1;
17                } else {
18                    dp[i][j] = Math.max(dp[i - 1][j], dp[i][j - 1]);
19                }
20            }
21        }
22        return dp[m][n];
23    }
24 }
```

☐ 4

☒ 0

☐ 1

☐ Error

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type
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Question No: 6

Multi Choice Type Question

What is the main drawback of using the dynamic programming approach to solve the Longest Increasing Subsequence (LIS) problem?

- ☐ High time complexity
- ☒ Requires additional memory ✓
- ☐ Only works for small arrays
- ☐ Does not produce accurate results

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Programming**
Blooms taxonomy **Analyse**

Question No: 7

Multi Choice Type Question

What will be the output of the following code?

```
1 class LCS {
2     public static void main(String[] args) {
3         String s1 = "ABCDE";
4         String s2 = "AEBDF";
5         int m = s1.length(), n = s2.length();
6         int[][] dp = new int[m+1][n+1];
7
8         for (int i = 0; i <= m; i++) {
9             for (int j = 0; j <= n; j++) {
10                if (i == 0 || j == 0)
11                    dp[i][j] = 0;
12                else if (s1.charAt(i-1) == s2.charAt(j-1))
13                    dp[i][j] = dp[i-1][j-1] + 1;
14                else
15                    dp[i][j] = Math.max(dp[i-1][j], dp[i][j-1]);
16            }
17        }
18        System.out.println(dp[5][4]);
19    }
20 }
```

- ☐ 4
- ☒ 3
- ☐ 5
- ☐ 6

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium**

Question No: 8

Multi Choice Type Question

Find the length of the longest increasing subsequence for the given sequence: $\{-10, 24, -9, 35, -21, 55, -41, 76, 84\}$

☐ 5

☐ 4

☐ 3

☒ 6



Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type	MCQ Single Correct	Subject
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Question No: 9

Multi Choice Type Question

What will be the output of the following code?

```
1 class LCS {
2     public static void main(String[] args) {
3         String s1 = "ABCDEFGF";
4         String s2 = "HIJKLMN";
5         System.out.println(lcs(s1, s2));
6         String s3 = "GFEDCBA";
7         String s4 = "ABCDEFGF";
8         System.out.println(lcs(s3, s4));
9     }
10
11     static int lcs(String s1, String s2) {
12         int m = s1.length();
13         int n = s2.length();
14         int[][] dp = new int[m + 1][n + 1];
15
16         for (int i = 1; i <= m; i++) {
17             for (int j = 1; j <= n; j++) {
18                 if (s1.charAt(i - 1) == s2.charAt(j - 1)) {
19                     dp[i][j] = dp[i - 1][j - 1] + 1;
20                 } else {
21                     dp[i][j] = Math.max(dp[i - 1][j], dp[i][j - 1]);
22                 }
23             }
24         }
25         return dp[m][n];
26     }
27 }
```

☒ 0
1

☐ 0
0

☐ 1
1

☐ 0
7

Status **Correct** Mark obtained **1/1** Hints used **0** Level **Medium** Question type

Question No: 10

Multi Choice Type Question

What will be the output of the following code?

```
1 public class LCS {
2     public static void main(String[] args) {
3         String s1 = "1234567";
4         String s2 = "7654321";
5         System.out.println(lcs(s1, s2));
6     }
7
8     static int lcs(String s1, String s2) {
9         int m = s1.length();
10        int n = s2.length();
11        int[][] dp = new int[m + 1][n + 1];
12
13        for (int i = 1; i <= m; i++) {
14            for (int j = 1; j <= n; j++) {
15                if (s1.charAt(i - 1) == s2.charAt(j - 1)) {
16                    dp[i][j] = dp[i - 1][j - 1] + 1;
17                } else {
18                    dp[i][j] = Math.max(dp[i - 1][j], dp[i][j - 1]);
19                }
20            }
21        }
22        return dp[m][n];
23    }
24 }
```

☒ 4

☐ 3

☐ 5

☐ 6

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type

Multi Choice Type Question

What will be the output of the following code?

```
1 class LCS {
2     public static void main(String[] args) {
3         String s1 = "ABCDEF";
4         String s2 = "FBDAMN";
5         int m = s1.length(), n = s2.length();
6         int[][] dp = new int[m+1][n+1];
7
8         for (int i = 0; i <= m; i++) {
9             for (int j = 0; j <= n; j++) {
10                 if (i == 0 || j == 0)
11                     dp[i][j] = 0;
12                 else if (s1.charAt(i-1) == s2.charAt(j-1))
13                     dp[i][j] = dp[i-1][j-1] + 1;
14                 else
15                     dp[i][j] = Math.max(dp[i-1][j], dp[i][j-1]);
16             }
17         }
18         System.out.println(dp[m][n]);
19     }
20 }
```

☐ 4☐ 3☐ 5☒ 2

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type
--------	---------	---------------	-----	------------	---	-------	--------	---------------

Question No: 12

Multi Choice Type Question

What is the time complexity of the dynamic programming solution for LCS?

☐ $O(2^n)$

☐ $O(n^2)$

☒ $O(m \times n)$

☐ $O(n)$

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Medium**

Question No: 13

Multi Choice Type Question

The number of increasing subsequences with the longest length for the given sequence {10, 9, 8, 7, 6, 5} are _____.

☒ 6

☐ 3

☐ 4

☐ 5

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Easy**

Question type **MCQ Single Correct**

Subject **Programming**

Question No: 14

Multi Choice Type Question

What is the space complexity of the standard DP solution for LCS?

☐ $O(n)$

☐ $O(m)$

☒ $O(m \times n)$

☐ $O(1)$

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium**

Question No: 15

Multi Choice Type Question

In the LCS DP table, what does $dp[i][j]$ represent?

☐ Maximum length of any subsequence

☐ Number of common characters at positions i and j

☐ Length of LCS of strings starting from position i and j

☒ Length of LCS of first i characters of string1 and first j characters of string2

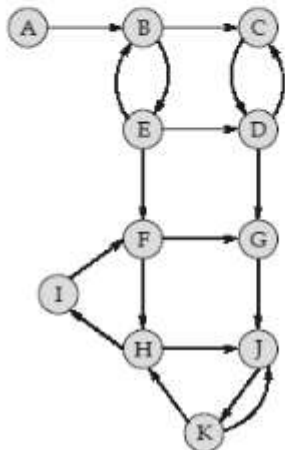
Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type

Lab 8 – Digraph

Question No: 1

Multi Choice Type Question

Consider the following graph:



The number of strongly connected components of the graph are

☐ 5

☒ 4

☐ 2

☐ 3

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Medium**

Question No: 2

Multi Choice Type Question

What will be the output of the following code?

```
1 public class PrintEdges {  
2     public static void main(String[] args) {  
3         int[][] edges = {{0,1},{1,2},{2,0}};  
4         for(int i=0;i<edges.length;i++) {  
5             System.out.println(edges[i][0] + " -> " + edges[i][1]);  
6         }  
7     }  
8 }  
9
```

☐ 1->0, 2->1, 0->2

☒ 0->1, 1->2, 2->0

☐ 0->2, 2->1, 1->0

☐ Error

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type

Multi Choice Type Question

Which of the following statements is true about strongly connected components in a digraph?

- ☐ Every strongly connected component has exactly one vertex
- ☐ A digraph with no cycles cannot have strongly connected components
- ☒ The strongly connected components of a digraph are disjoint
- ☐ A digraph with a single strongly connected component is a tree

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type	MCQ Single Correct
--------	----------------	---------------	------------	------------	----------	-------	---------------	---------------	---------------------------

Multi Choice Type Question

What is the output of the strongly connected components in the given digraph with multiple components?

```

1 import java.util.ArrayList;
2 import java.util.List;
3 class StronglyConnectedComponents {
4     public static void main(String[] args) {
5         int[][] graph = {
6             {0, 1, 0, 0, 0, 0},
7             {0, 0, 1, 0, 0, 0},
8             {0, 0, 0, 1, 0, 0},
9             {1, 0, 0, 0, 0, 0},
10            {0, 0, 0, 0, 0, 1},
11            {0, 0, 0, 0, 1, 0}
12        };
13        List<List<Integer>> scc = findStronglyConnectedComponents(graph);
14        System.out.println(scc);
15    }
16    public static List<List<Integer>> findStronglyConnectedComponents(int[][] graph) {
17        int n = graph.length;
18        boolean[] visited = new boolean[n];
19        List<List<Integer>> scc = new ArrayList<>();
20        for (int i = 0; i < n; i++) {
21            if (!visited[i]) {
22                List<Integer> component = new ArrayList<>();
23                dfs(graph, i, visited, component);
24                scc.add(component);
25            }
26        }
27        return scc;
28    }
29    public static void dfs(int[][] graph, int vertex, boolean[] visited, List<Integer> component) {
30        visited[vertex] = true;
31        component.add(vertex);
32        for (int i = 0; i < graph.length; i++) {
33            if (graph[vertex][i] == 1 && !visited[i]) {
34                dfs(graph, i, visited, component);
35            }
36        }
37    }
38 }

```

☒ `[[0, 1, 2, 3], [4, 5]]`

☐ `[[0], [1], [2], [3], [4, 5]]`

☐ `[[0, 1, 2, 3], [4], [5]]`

☐ `[[0, 1, 2, 3], [4], [5]]`

Question No: 5

Multi Choice Type Question

What will be the output of the following code?

```
1 class StronglyConnectedGraph {
2     public static void main(String[] args) {
3         int[][] graph = {
4             {0, 1, 0, 0},
5             {0, 0, 1, 0},
6             {0, 0, 0, 1},
7             {1, 0, 0, 0}
8         };
9
10        if (isStronglyConnected(graph)) {
11            System.out.println("The graph is strongly connected.");
12        } else {
13            System.out.println("The graph is NOT strongly connected.");
14        }
15    }
16
17    public static boolean isStronglyConnected(int[][] graph) {
18        int n = graph.length;
19        for (int start = 0; start < n; start++) {
20            boolean[] visited = new boolean[n];
21            dfs(graph, start, visited);
22            for (boolean v : visited) {
23                if (!v) return false;
24            }
25        }
26        return true;
27    }
28
29    public static void dfs(int[][] graph, int vertex, boolean[] visited) {
30        visited[vertex] = true;
31        for (int i = 0; i < graph.length; i++) {
32            if (graph[vertex][i] == 1 && !visited[i]) {
33                dfs(graph, i, visited);
34            }
35        }
36    }
37 }
38 }
```

☐ The graph is NOT strongly connected.

☒ The graph is strongly connected.

☐ Runtime error

☐ Compilation error

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Question No: 6

Multi Choice Type Question

Consider a digraph with n vertices, what is the time complexity of Tarjan's algorithm for finding strongly connected components?

☐ $O(n^3)$

☐ $O(n^2)$

☒ $O(n + m)$

☐ $O(n!)$

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Data Structure**

Question No: 7

Multi Choice Type Question

In the context of strongly connected components, what is the relationship between a vertex and its low-link value in Tarjan's algorithm?

☒ The low-link value is the smallest index reachable from the vertex ✓

☐ The low-link value is the discovery time of the vertex

☐ The low-link value is the finishing time of the vertex

☐ The low-link value is the largest index reachable from the vertex

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Data Structure**

Question No: 8

Multi Choice Type Question

Suppose we are given a digraph and asked to determine if it is strongly connected. However, instead of using an algorithm for strongly connected components, we decide to check for the existence of a path from every vertex to every other vertex. What is the time complexity of this approach?

☐ $O(n + m)$

☐ $O(n^2 + m)$

☐ $O(n^3)$

☒ $O(n * (n + m))$ ✓

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Data Structure** | Topic **Graph** | Sub Topic **Graph** | Blooms taxonomy **Understand**

Question No: 9

Multi Choice Type Question

In a digraph, if we have two strongly connected components, A and B, and there is an edge from a vertex in A to a vertex in B, what can we conclude about the strongly connected components?

☐ A and B will always merge into a single strongly connected component

☒ A and B will remain separate strongly connected components ✓

☐ There will be a path from every vertex in B to every vertex in A

☐ There will be no path from any vertex in B to any vertex in A

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Data Structure** | Topic **Graph** | Sub Topic **Graph** | Blooms taxonomy **Understand**

Multi Choice Type Question

What will be the output of the following code?

```

1 class StronglyConnectedGraph {
2     public static void main(String[] args) {
3         int[][] graph = {
4             {0, 1, 0, 0, 0},
5             {0, 0, 1, 0, 0},
6             {0, 0, 0, 1, 0},
7             {0, 0, 0, 0, 1},
8             {0, 0, 1, 0, 0}
9         };
10
11         if (isStronglyConnected(graph)) {
12             System.out.println("The graph IS strongly connected.");
13         } else {
14             System.out.println("The graph is NOT strongly connected.");
15         }
16     }
17
18     public static boolean isStronglyConnected(int[][] graph) {
19         int n = graph.length;
20         for (int start = 0; start < n; start++) {
21             boolean[] visited = new boolean[n];
22             dfs(graph, start, visited);
23             for (boolean v : visited) {
24                 if (!v) return false;
25             }
26         }
27         return true;
28     }
29
30     public static void dfs(int[][] graph, int vertex, boolean[] visited) {
31         visited[vertex] = true;
32         for (int i = 0; i < graph.length; i++) {
33             if (graph[vertex][i] == 1 && !visited[i]) {
34                 dfs(graph, i, visited);
35             }
36         }
37     }
38 }
39

```

☒ The graph is NOT strongly connected.

☐ Runtime error

☐ The graph IS strongly connected.

☐ Compilation error

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Easy	Question type
--------	---------	---------------	-----	------------	---	-------	------	---------------

Question No: 11

Multi Choice Type Question

What is the output of the strongly connected components in the given digraph?

```
1 import java.util.ArrayList;
2 import java.util.List;
3 public class StronglyConnectedComponents {
4     public static void main(String[] args) {
5         int[][] graph = {
6             {0, 1, 0, 0},
7             {0, 0, 1, 0},
8             {0, 0, 0, 1},
9             {1, 0, 0, 0}
10        };
11        List<List<Integer>> scc = findStronglyConnectedComponents(graph);
12        System.out.println(scc);
13    }
14    public static List<List<Integer>> findStronglyConnectedComponents(int[][] graph) {
15        int n = graph.length;
16        boolean[] visited = new boolean[n];
17        List<List<Integer>> scc = new ArrayList<>();
18        for (int i = 0; i < n; i++) {
19            if (!visited[i]) {
20                List<Integer> component = new ArrayList<>();
21                dfs(graph, i, visited, component);
22                scc.add(component);
23            }
24        }
25        return scc;
26    }
27    public static void dfs(int[][] graph, int vertex, boolean[] visited, List<Integer> component) {
28        visited[vertex] = true;
29        component.add(vertex);
30        for (int i = 0; i < graph.length; i++) {
31            if (graph[vertex][i] == 1 && !visited[i]) {
32                dfs(graph, i, visited, component);
33            }
34        }
35    }
36 }
```

☐ [[0], [1, 2, 3]]

☐ [[0, 1, 2], [3]]

☒ [[0, 1, 2, 3]]

Question No: 12

Multi Choice Type Question

In a digraph, a strongly connected component is a subgraph where there is a path from every vertex to every other vertex. Which of the following algorithms is most efficient for finding strongly connected components in a digraph with a large number of vertices and edges?

☐ Dijkstra's algorithm

☐ Kosaraju's algorithm

☐ Bellman-Ford algorithm

☒ Tarjan's algorithm

Status: Correct | Mark obtained: 1/1 | Hints used: 0 | Level: Medium | Question type: MCQ Single Correct | Subject: Data Structure | Topic: Graph | Sub Topic: Graph | Blooms taxonomy: Understand

Question No: 13

Multi Choice Type Question

A graph having an edge from each vertex to every other vertex is called a _____.

☒ Tightly Connected

☐ Strongly Connected

☐ Weakly Connected

☐ Loosely Connected

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Medium**

Question type

Question No: 14

Multi Choice Type Question

A digraph is said to be strongly connected if it has exactly one strongly connected component. Consider a digraph with vertices $\{1, 2, 3, 4\}$ and edges $\{(1, 2), (2, 3), (3, 1), (3, 4)\}$. Is this digraph strongly connected?

☐ Yes

☐ Maybe

☒ No ✓

☐ Cannot be determined

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Medium**

Question type **MCQ Single Correct**

Subject **Data Structure**

Topic **Graph**

Sub Topic **Graph**

Blooms taxonomy **Understand**

Question No: 15

Multi Choice Type Question

Let $G = (V, E)$ be a directed graph where V is the set of vertices and E the set of edges. Then which one of the following graphs has the same strongly connected component

☐ $G_1 = (V_1, E_1)$ where V_1 is the set of vertices in G which are not isolated

☐ where $E = \{(u, v) | (u, v) \in E\}$

☐ $G_3 = (V, E_3)$ where $E_3 = \{(u, v) | \text{there is a path of length } \leq 2 \text{ from } u \text{ to } v \text{ in } E\}$

☒ $G_2 = (V, E_2)$ where $E_2 = \{(u, v) | (v, u) \in E\}$ ✓

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Data Structure** | Topic **Graph** | Sub Topic **Graph**

Lab 9 – APSP

Question No: 1

Multi Choice Type Question

Floyd- Warshall algorithm was proposed by _____

☒ Robert Floyd and Stephen Warshall

☐ Bernad Floyd and Robert Warshall

☐ Robert Floyd and Bernad Warshall

☐ Stephen Floyd and Robert Warshall

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium**

Question No: 2

Multi Choice Type Question

Which of the following algorithms can find the shortest path between all pairs of nodes in a graph?

☐ Backtracking algorithm

☐ Greedy algorithm

☒ Floyd-Warshall algorithm



☐ Dijkstra's algorithm

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 3

Multi Choice Type Question

In the Floyd Warshall algorithm, how many iterations are required to find the shortest path between all pairs of vertices in a graph with N vertices?

☐ N+1

☐ N-1

☐ N^2

☒ N



Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Algorithms** | Topic **Graph**

Question No: 4

Multi Choice Type Question

What is the time complexity of the Floyd Warshall Algorithm, where V is the number of vertices in the graph?

☐ $O(V^2)$

☐ $O(V)$

☐ None of the mentioned options

☒ $O(V^3)$



Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 5

Multi Choice Type Question

Which data structure is often used to implement the Floyd-Warshall algorithm?

☐ Hash Table

☐ None of the mentioned options

☐ Linked List

☒ Array

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type

Question No: 6

Multi Choice Type Question

Floyd Warshall Algorithm can be used for finding _____.

- ☐ Minimum spanning tree
- ☒ Transitive closure
- ☐ Single source shortest path
- ☐ Topological sort

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Question No: 7

Multi Choice Type Question

Which of the following is a disadvantage of using Floyd-Warshall algorithm compared to Dijkstra's algorithm for finding the shortest path in a graph?

- ☐ Cannot handle negative edge weights
- ☐ Only applicable to unweighted graphs
- ☒ Higher time complexity ✓
- ☐ Only applicable to directed graphs

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Easy**

Question type **MCQ Single Correct**

Subject

Algorithms

Topic

Graph

Question No: 8

Multi Choice Type Question

Which of the following algorithms solves the single-source shortest paths?

☒ Dijkstra's Algorithm

☐ Krushkal's Algorithm

☐ Floyd-Warshall Algorithm

☐ Prims Algorithm

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Medium**

Question No: 9

Multi Choice Type Question

What happens when the value of k is 0 in the Floyd Warshall Algorithm?

- ☒ 0 intermediate vertex
- ☐ N intermediate vertices
- ☐ 1 intermediate vertex
- ☐ N-1 intermediate vertices

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Easy**

Question No: 10

Multi Choice Type Question

A search algorithm is an example of which type of shortest path algorithm?

☐ Dynamic programming algorithm

☐ Greedy algorithm

☐ Dijkstra's algorithm

☒ Heuristic algorithm

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Medium**

Question No: 11

Multi Choice Type Question

In the context of shortest path algorithms, what does "edge weight" represent?

- ☐ The number of nodes in the graph
- ☒ The cost or distance associated with traversing an edge
- ☐ The physical length of an edge in a graph
- ☐ The number of neighbors a node has

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type

Question No: 12

Multi Choice Type Question

Which of the following is a critical difference between Dijkstra's algorithm and the Floyd-Warshall algorithm?

- ☐ Dijkstra's algorithm can handle graphs with negative edge weights.
- ☒ Dijkstra's algorithm always finds the shortest path to all nodes. ✓
- ☐ The Floyd-Warshall algorithm is a single-source algorithm.
- ☐ The Floyd-Warshall algorithm uses a priority queue.

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 13

Multi Choice Type Question

What is the main objective of a shortest-path algorithm?

- ☒ To find the most efficient path between two points in a graph
- ☐ To find the path with the fewest edges in a graph
- ☐ None of the mentioned options
- ☐ To find the path with the highest edge weights in a graph

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium**

Question No: 14

Multi Choice Type Question

In the Floyd-Warshall algorithm, what does it mean if the path matrix contains a value ∞ for a pair of nodes?

- ☐ The path is of infinite length
- ☒ There is no path between the nodes ✓
- ☐ The nodes have a direct edge with weight ∞
- ☐ None of the mentioned options

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Question No: 15

Multi Choice Type Question

What is the time complexity of the Floyd-Warshall algorithm for finding the all pairs shortest path in a graph with n nodes?

☒ $O(n^3)$



☐ $O(n \log n)$

☐ $O(n)$

☐ $O(n^2)$

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Easy	Question type	MCQ Single Correct	Subject	Algorithms
--------	----------------	---------------	------------	------------	----------	-------	-------------	---------------	---------------------------	---------	-------------------

Lab 10 – Maximum Flow

Question No: 1

Multi Choice Type Question

In the iterative improvement method for the maximum-flow problem, the residual graph is updated after each augmentation. What is the purpose of updating the residual graph?

- ☐ To increase the capacity of the edges in the network
- ☐ To minimize the flow in the network
- ☐ To reduce the capacity of the edges in the network
- ☒ To reflect the changes in the flow values and edge capacities ✓

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type	MCQ Single Correct	Subject	Algorithms	Topic	Maximum Flow	Sub Topic	Limitations of Algorithms
Blooms taxonomy	Understand														

Question No: 2

Multi Choice Type Question

In the iterative improvement method for the maximum-flow problem, the flow value is updated along the augmenting path by the minimum capacity of the edges in the path. What is the effect of this update on the residual graph?

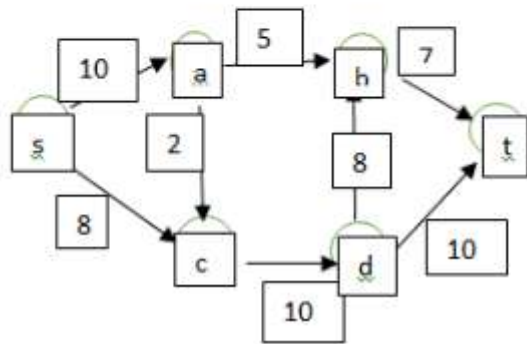
- ☐ The residual capacities of the edges in the augmenting path are reset to their original values
- ☐ The residual capacities of the edges in the augmenting path are increased
- ☐ The residual capacities of the edges in the augmenting path are unchanged
- ☒ The residual capacities of the edges in the augmenting path are decreased

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type	MCQ Single Correct	Subject	Algorithms	Topic	Maximum Flow	Sub Topic	Limitations of Algorithms
Bloms taxonomy	Understand														

Question No: 3

Multi Choice Type Question

Find the maximum flow from the following graph.



☐ 22

☒ 15

☐ 17

☐ 20

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Question No: 4

Multi Choice Type Question

How many constraints does flow have?

☐ Three

☒ Two

☐ One

☐ Four

Status **Correct** | Mark obtained **1/1**

Question No: 5

Multi Choice Type Question

In the iterative improvement method for the maximum-flow problem, what is the condition for terminating the algorithm?

☐ When the flow value reaches the maximum possible flow

☐ When the number of iterations exceeds a predefined limit

☒ When there are no more augmenting paths in the residual graph ✓

☐ When the residual capacity of all edges becomes zero

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject

Question No: 6

Multi Choice Type Question

What is the time complexity of the iterative improvement method for the maximum-flow problem in the worst case?

☐ $O(V^3E^2)$

☐ $O(VE)$

☒ $O(VE^2)$ ✓

☐ $O(V^2E)$

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject

Question No: 7

Multi Choice Type Question

How does the Iterative Improvement Method handle the issue of negative-weight edges in the residual graph?

☒ It is not designed to handle negative-weight edges ✓

☐ It inverts the sign of negative-weight edges

☐ It uses a special case to handle negative-weight edges

☐ It ignores negative-weight edges

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject

Question No: 8

Multi Choice Type Question

Suppose we have a flow network with a source node s and a sink node t , and we want to find the maximum flow from s to t . If the capacity of the edge from s to t is 10, and there is no other path from s to t , what is the maximum flow using the iterative improvement method?

☐ 5

☐ 20

☐ 0

☒ 10 ✓

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct** | Subject **Algorithms** | Topic **Maximum Flow** | Sub Topic **Limitations of Algorithms** | Blooms taxonomy **Understand**

Question No: 9

Multi Choice Type Question

What is the source?

- ☐ Vertex with the least weight
- ☐ Centre vertex
- ☐ Vertex with no leaving edges
- ☒ Vertex with no incoming edges

Status **Correct** | Mark obtained **1/1**

Question No: 10

Multi Choice Type Question

What does the Maximum flow problem involve?

- ☐ Computing a minimum-spanning tree
- ☒ Finding a flow between source and sink that is maximum
- ☐ Finding a flow between source and sink that is minimum
- ☐ Finding the shortest path between source and sink

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Easy**

Question No: 11

Multi Choice Type Question

In the iterative improvement method for the maximum-flow problem, the flow value is updated along the augmenting path. What is the flow value updated by?

- ☐ The maximum capacity of the edges in the augmenting path
- ☐ The sum of the capacities of the edges in the augmenting path
- ☒ The minimum capacity of the edges in the augmenting path ✓
- ☐ The difference between the capacities of the edges in the augmenting path

Status **Correct**

Mark obtained **1/1**

Hints used **0**

Level **Medium**

Question type **MCQ Single Correct**

Subject **Algorithms**

Topic **Maximum Flow**

Question No: 12

Multi Choice Type Question

A simple acyclic path between source and sink that passes through only positive-weighted edges is called

- ☐ Critical path
- ☐ Residual path
- ☐ Maximum path
- ☒ Augmenting path

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Easy** | Question type **MCQ Single Correct**

Question No: 13

Multi Choice Type Question

In the Iterative Improvement Method, what is the purpose of finding augmenting paths in the residual graph?

- ☐ To reduce the flow in the original graph
- ☐ To determine the maximum flow in the flow network
- ☒ To increase the flow in the original graph
- ☐ To identify the minimum cut in the flow network

Status **Correct** | Mark obtained **1/1** | Hints used **0** | Level **Medium** | Question type **MCQ Single Correct**

Question No: 14

Multi Choice Type Question

In the context of the maximum-flow problem, what is the purpose of the residual graph?

- ☐ To find the shortest path in the original flow network
- ☐ To keep track of the minimum cut in the flow network
- ☒ To identify the augmenting paths in the flow network
- ☐ To keep track of the maximum flow in the flow network

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type
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Question No: 15

Multi Choice Type Question

Under what conditions can a vertex combine and distribute flow in any manner?

- ☐ The vertex should be a sink vertex
- ☐ It may violate edge capacities
- ☐ The vertex should be a source vertex
- ☒ It should maintain flow conservation

Status	Correct	Mark obtained	1/1	Hints used	0	Level	Medium	Question type
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